

Homework 3

Repository: <https://github.com/Ani2508/data260-3209>

Domain: Clinical Trial Listing

Personal Configuration

Value	Result
SID4	3209
PORT_BASE	8509 (8000 + 3209 mod 900)
PREFIX	s3209
SEED	3209
VERIFY_SEED	263209
DOMAIN_ID	1 -Clinical trial listings

Environment

Hardware	Microsoft Surface Laptop 3 — AMD Ryzen 5 (4 cores), integrated AMD Radeon (no GPU acceleration)
RAM	16GB
OS	Windows 11 Home 64-bit
Embedding model:	sentence-transformers/all-MiniLM-L6-v2

Part 1 FastAPI Authentication Application

1. Routes

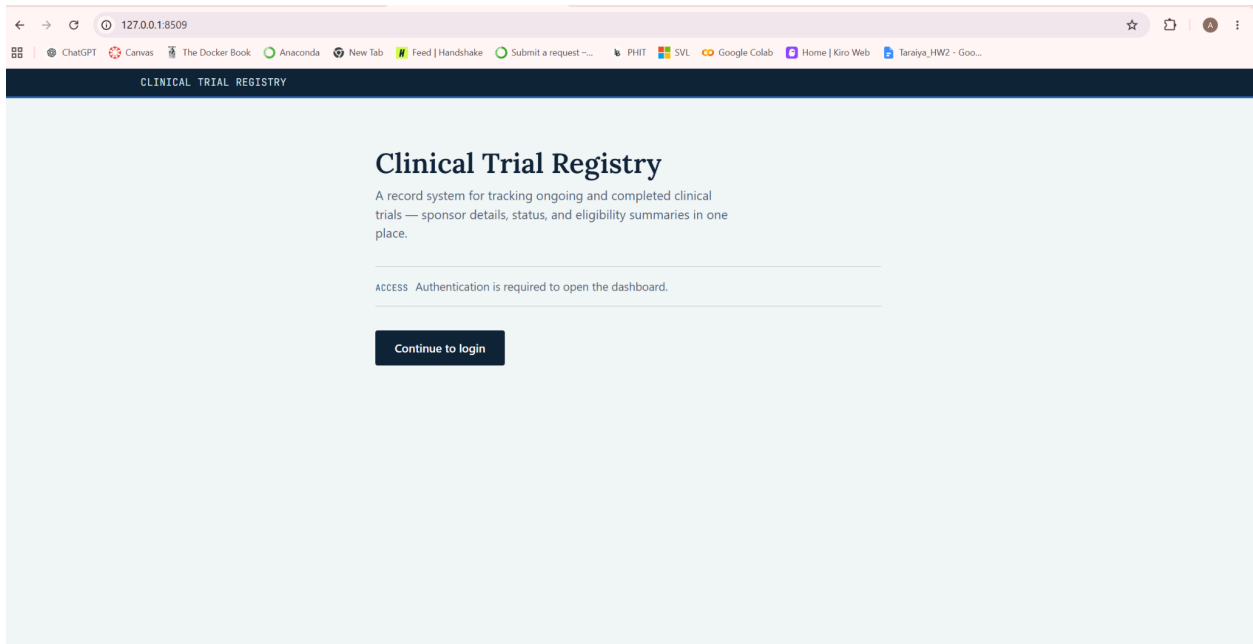
The application implements four authentication routes for the Clinical Trial Registry: home, login, dashboard and logout. The routes are placed in the separate auth.py router as required.

Home Route

Code Snippet

```
33
34 @router.get("/", response_class=HTMLResponse)
35 def home(request: Request):
36     username = request.session.get("user")
37
38     if username and session_is_active(request):
39         return templates.TemplateResponse(
40             request=request,
41             name="home.html",
42             context={"username": username},
43         )
44
45     request.session.clear()
46
47     return templates.TemplateResponse(
48         request=request,
49         name="home.html",
50         context={"username": None},
51     )
52
```

Output Snippet:



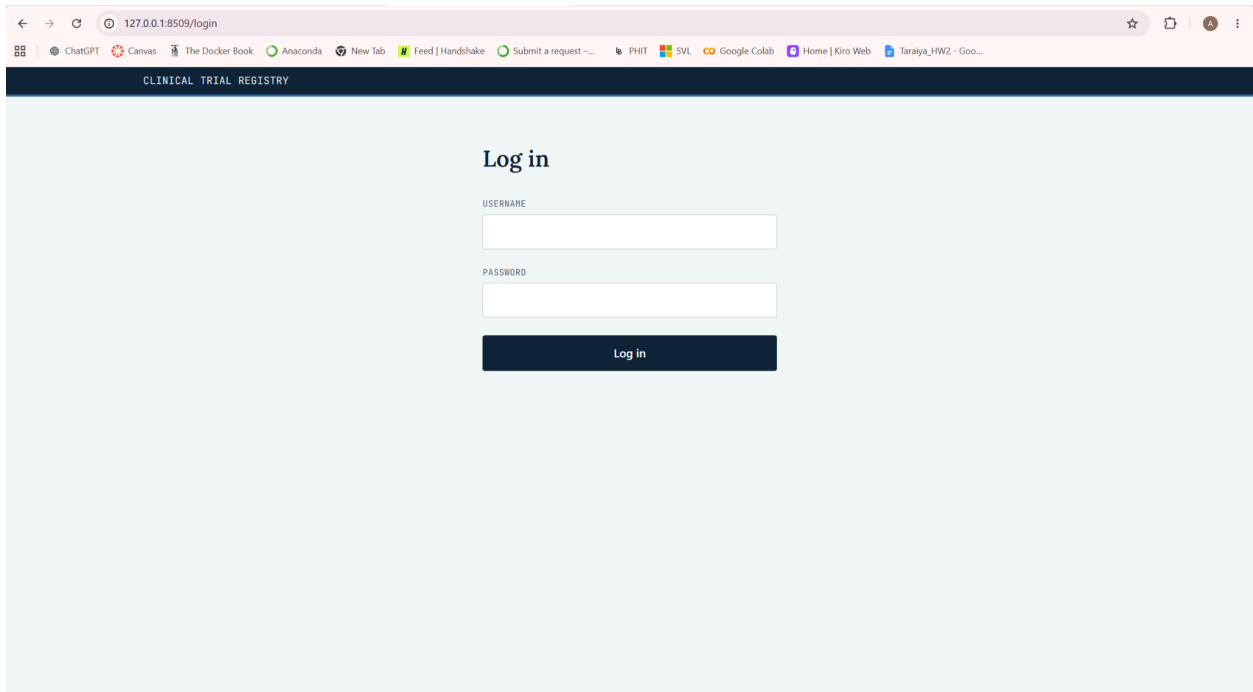
The home page displays the Clinical Trial Registry welcome message. When the user is logged out it displays a link to the login page

Login route

Code Snippet

```
64 @router.get("/login", response_class=HTMLResponse)
65 def login_page(request: Request):
66     return templates.TemplateResponse(
67         request=request,
68         name="login.html",
69         context={"message": None},
70     )
71
72
73 @router.post("/login", response_class=HTMLResponse)
74 def login(
75     request: Request,
76     username: str = Form(...),
77     password: str = Form(...),
78 ):
79     if username != DEMO_USERNAME or password != DEMO_PASSWORD:
80         return templates.TemplateResponse(
81             request=request,
82             name="login.html",
83             context={"message": "Invalid username or password."},
84             status_code=401,
85         )
86
87     request.session.clear()
88     request.session["user"] = username
89     request.session["last_activity"] = time.time()
90
91     return RedirectResponse(url="/dashboard", status_code=303)
92
```

Output Snippet:



The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8509/login'. The browser's tab bar includes several open tabs: ChatGPT, Canvas, The Docker Book, Anaconda, New Tab, Feed | Handshake, Submit a request..., PHIT, SVL, Google Colab, Home | Kiro Web, and Taralya_HW2 - Goo... The page title is 'CLINICAL TRIAL REGISTRY'. The main content area features a 'Log in' heading, followed by two input fields labeled 'USERNAME' and 'PASSWORD'. Below these fields is a dark blue 'Log in' button.

The login form accepts a username and password. Invalid credentials return a Bootstrap alert with status 401. Correct credentials create a session and redirect the user to the dashboard.

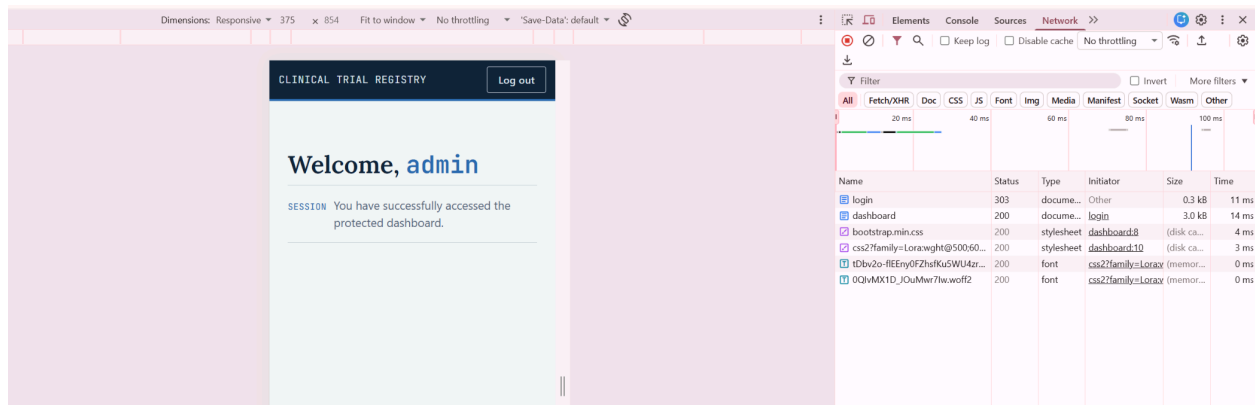
Dashboard route

Code snippet:

```
@router.get("/dashboard", response_class=HTMLResponse)
def dashboard(request: Request):
    if not session_is_active(request):
        return RedirectResponse(url="/login", status_code=303)

    return templates.TemplateResponse(
        request=request,
        name="dashboard.html",
        context={"username": request.session["user"]},
    )
```

Output Snippet



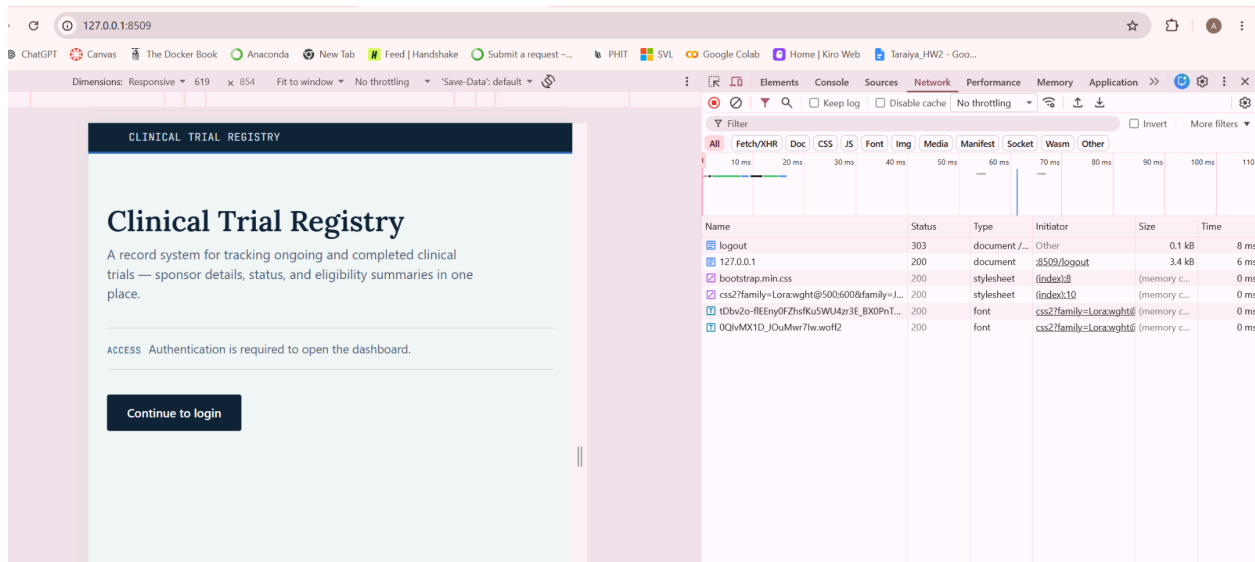
The dashboard is protected. Only users with an active session can access it. The page displays the authenticated user's name.

Logout route

Code snippet:

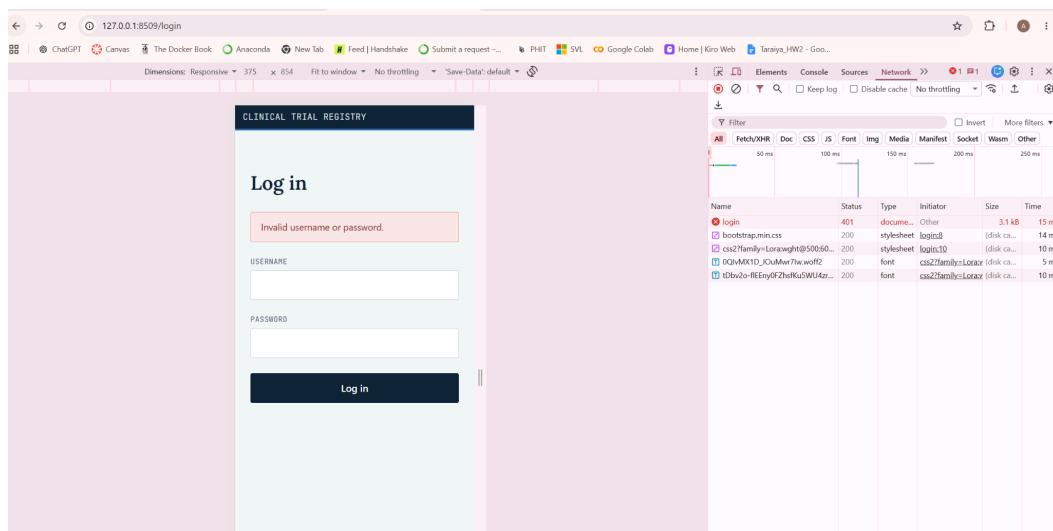
```
94  
95  
96 @router.get("/logout")  
97 def logout(request: Request):  
98     request.session.clear()  
99     return RedirectResponse(url="/", status_code=303)
```

Output Snippet:



The logout route clears the session and redirects the user to the home page. Authenticated dashboard showing the logged-in user's name. The Network tab confirms that login redirected with status 303 and the dashboard returned status 200.

Test Invalid Credentials



2. Session Management

Code snippet:

```

15
16 app = FastAPI(title="Clinical Trial Listing API")
17
18 app.add_middleware(
19     SessionMiddleware,
20     secret_key="hw3-session-secret-3209",
21     max_age=300,
22     same_site="lax",
23     https_only=True,
24 )
25
26 app.include_router(auth_router)
27
28 BASE_DIR = Path(__file__).resolve().parent
29

```

Starlette's SessionMiddleware manages the user session. The session cookie is configured with a five-minute maximum age, SameSite protection, and Secure cookie behavior.

```

request.session.clear()
request.session["user"] = username
request.session["last_activity"] = time.time()

```

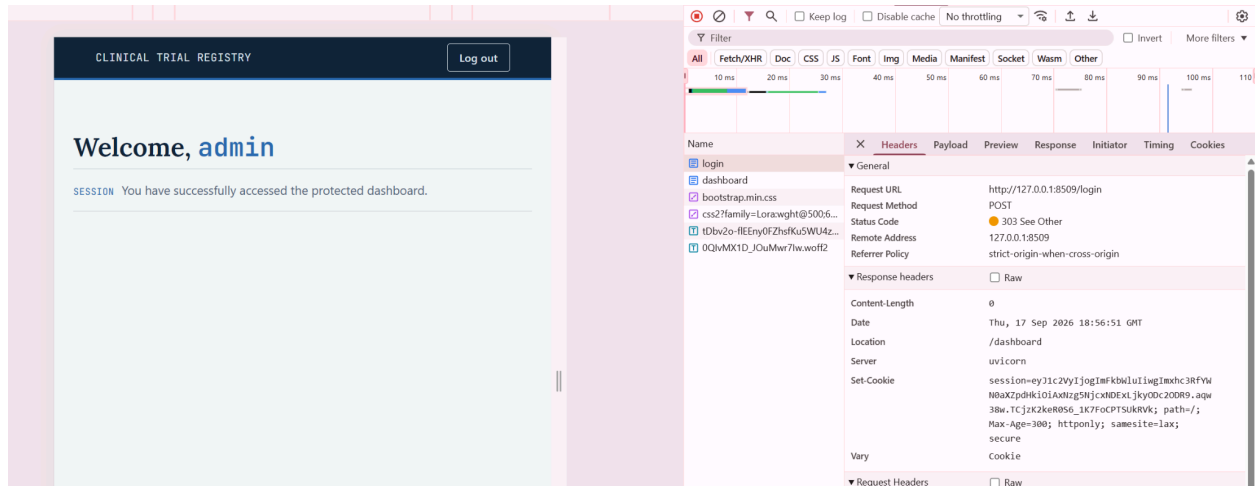
After successful login, the username and the last activity time are stored in the session.

```

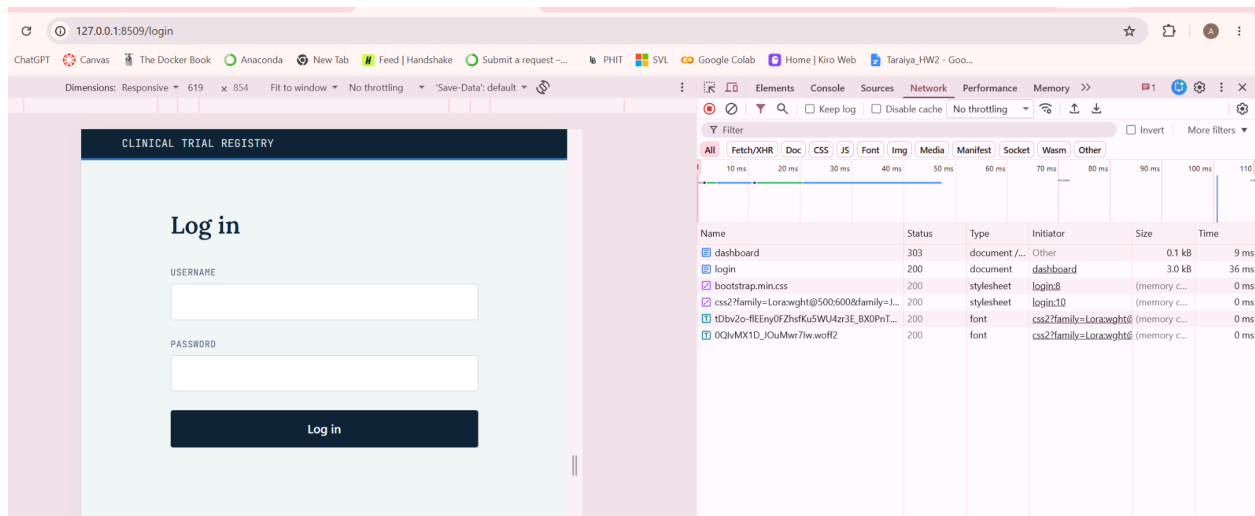
if time.time() - last_activity > SESSION_TIMEOUT_SECONDS:
    request.session.clear()
    return False

```

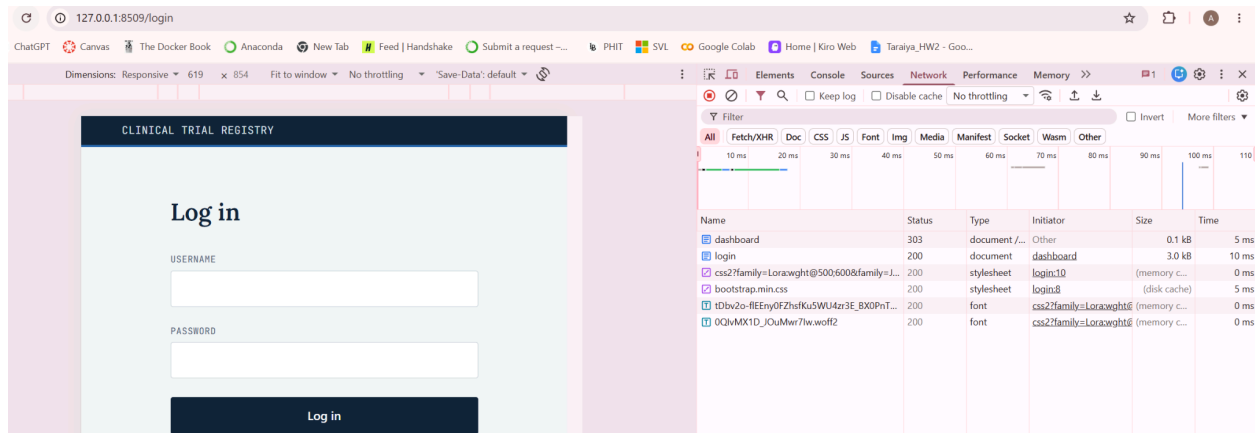
Output Snippet:



Browser Network inspection of the successful login response. The response redirects to `/dashboard` and sets a session cookie with Max-Age, HttpOnly, SameSite, and Secure attributes.



After logout, attempting to open `/dashboard` produces a 303 redirect to `/login`. This demonstrates that a logged-out user cannot access the protected dashboard.



After the session remained inactive for 300 seconds, a new request to /dashboard returned status 303 and redirected the user to /login. This demonstrates the idle session timeout.

3. Bootstrap Styling

Code snippet:

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?family=Lora:wght@500;600&family=JetBrains+Mono:wght@400;600&display=swap" rel="stylesheet">
```

```
<body>
  <nav class="navbar registry-bar">
    <div class="container">
      <span class="brand">CLINICAL TRIAL REGISTRY</span>
    </div>
  </nav>

  <main>
    <h1 class="file-heading">Clinical Trial Registry</h1>
    <p class="file-sub">
      A record system for tracking ongoing and completed clinical trials –
      sponsor details, status, and eligibility summaries in one place.
    </p>

    <div class="status-line">
      <span class="status-label">ACCESS</span>
      <span class="status-value">Authentication is required to open the dashboard.</span>
    </div>

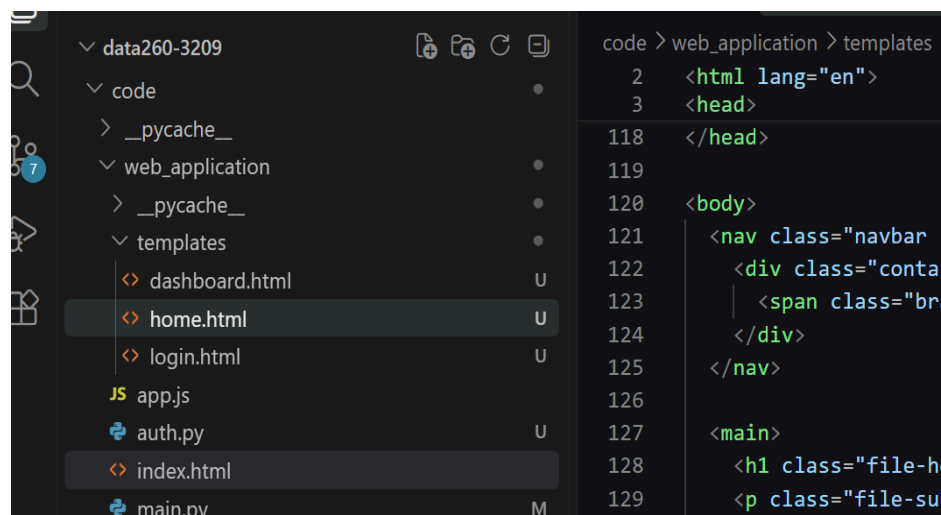
    {% if username %}
      <p class="signed-in-as">Signed in as <strong>{{ username }}</strong></p>
      <a href="/dashboard" class="btn btn-ledger me-2">Open dashboard</a>
      <a href="/logout" class="btn btn-ledger-outline">Log out</a>
    {% else %}
      <a href="/login" class="btn btn-ledger">Continue to login</a>
    {% endif %}
  </main>
</body>
```

Output Debug Console Terminal Ports

Cookie: session=null; path=/; expires=Thu, 01 Jan 1970 00:00:00 INFO: 127.0.0.1:57459 - "GET /dashboard HT

Bootstrap was used throughout the authentication interface. Navbar, cards, buttons, and alerts were used to create a consistent and responsive design for all pages.

Templates directory



```
code > web_application > templates
2  <html lang="en">
3  <head>
118 </head>
119
120 <body>
121   <nav class="navbar
122     <div class="conta
123       <span class="br
124     </div>
125   </nav>
126
127   <main>
128     <h1 class="file-h
129     <p class="file-su
```

Templates directory containing the HTML files used for the home, login, and dashboard pages.

Appendix A — [auth.py](#)

```
import time

from pathlib import Path

from fastapi import APIRouter, Form, Request
from fastapi.responses import HTMLResponse, RedirectResponse
from fastapi.templating import Jinja2Templates

router = APIRouter()

BASE_DIR = Path(__file__).resolve().parent
templates = Jinja2Templates(directory=str(BASE_DIR / "templates"))

DEMO_USERNAME = "admin"
DEMO_PASSWORD = "hw3pass"
SESSION_TIMEOUT_SECONDS = 300

def session_is_active(request: Request) -> bool:

    username = request.session.get("user")

    last_activity = request.session.get("last_activity")
```

```
if not username or not last_activity:
```

```
    return False
```

```
if time.time() - last_activity > SESSION_TIMEOUT_SECONDS:
```

```
    request.session.clear()
```

```
    return False
```

```
request.session["last_activity"] = time.time()
```

```
return True
```

```
@router.get("/", response_class=HTMLResponse)
```

```
def home(request: Request):
```

```
    username = request.session.get("user")
```

```
if username and session_is_active(request):
```

```
    return templates.TemplateResponse(
```

```
        request=request,
```

```
        name="home.html",
```

```
        context={"username": username},
```

```
    )
```

```
request.session.clear()

return templates.TemplateResponse(

    request=request,

    name="home.html",

    context={"username": None},

)

@router.get("/login", response_class=HTMLResponse)
def login_page(request: Request):

    return templates.TemplateResponse(

        request=request,

        name="login.html",

        context={"message": None},

    )

@router.post("/login", response_class=HTMLResponse)
def login(

    request: Request,

    username: str = Form(...),

    password: str = Form(...),
```

```

):

    if username != DEMO_USERNAME or password != DEMO_PASSWORD:

        return templates.TemplateResponse(

            request=request,

            name="login.html",

            context={"message": "Invalid username or password."},

            status_code=401,

        )

    request.session.clear()

    request.session["user"] = username

    request.session["last_activity"] = time.time()

    return RedirectResponse(url="/dashboard", status_code=303)

@router.get("/dashboard", response_class=HTMLResponse)
def dashboard(request: Request):

    if not session_is_active(request):

        return RedirectResponse(url="/login", status_code=303)

    return templates.TemplateResponse(

        request=request,

```

```

        name="dashboard.html",

        context={"username": request.session["user"]},

    )

@router.get("/logout")
def logout(request: Request):

    request.session.clear()

    return RedirectResponse(url="/", status_code=303)

```

Part 2: Retrieval-Only RAG Comparison

Objective

The objective of Part 2 is to compare three LlamaIndex chunking techniques for clinical-trial document retrieval. The same corpus, embedding model, questions, and retrieval settings are used for a fair comparison. This is retrieval-only RAG because the system retrieves relevant document chunks but does not generate final answers.

Environment & Setup Install

The required LlamaIndex and retrieval packages were installed. The public Hugging Face model `sentence-transformers/all-MiniLM-L6-v2` was downloaded and tested successfully. It converts text into 384-dimensional embedding vectors.

```

9   from llama_index.embeddings.huggingface import HuggingFaceEmbeddin
10
11
12   PROJECT_ROOT = Path(__file__).resolve().parents[3]
13   CORPUS_DIR = PROJECT_ROOT / "corpus"
14   EMBEDDING_MODEL_NAME = "sentence-transformers/all-MiniLM-L6-v2"
15

```

```
instrumentation-0.6.0 llama-index-llms-openai-0.7.10 llama-index-workflows-2.24.0
U markdown-it-py-4.2.0 marshmallow-3.26.2 mdurl-0.1.2 multidict-6.8.0 mypy-extensions-1.1.0
U narwhals-2.26.0 nest-asyncio-1.6.0 nltk-3.10.3 openai-2.54.0 propcache-0.5.4
regex-2026.9.10 rich-15.0.0 safetensors-0.8.0 scikit-learn-1.9.1 scipy-1.18.1
sentence-transformers-6.0.1 shellingham-1.5.4 sqlalchemy-2.0.54 threadpoolctl-3.7.0
tiktoken-0.14.0 tinytag-2.3.2 tokenizers-0.23.2 tqdm-4.70.1 transformers-5.17.0
typer-0.27.2 typing-inspect-0.9.0 wrapt-2.4.1 yarl-1.25.1

[notice] A new release of pip is available: 24.3.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\data260-3209> py -3.12 -c "import llama_index, llama_index.embeddings.huggingface,
sentence_transformers, faiss, numpy, pandas; print('All HW3 Part 2 packages are installed')"
```

Hugging Face is the public model repository where we download the embedding model. We are using: sentence-transformers/all-MiniLM-L6-v2, It converts text into a list of numbers called an embedding vector. This model creates a vector with 384 numbers. LlamaIndex uses these vectors to compare the meaning of the query with the meaning of document chunks.

```
KS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
model.safetensors: downloading bytes: █ 85.0MB, 7.50MB/
model.safetensors: reconstructing file: 100%██ 90.9MB /
Loading weights: 100%██ 103/103 [00:00<00:00, 1166.42it
tokenizer_config.json: 100%██ 350/350 [00:00<?, ?B/s]
vocab.txt: 100%██ 232k/232k [00:00<00:00, 22.4MB/s]
tokenizer.json: 100%██ 466k/466k [00:00<00:00, 156MB/s]
special_tokens_map.json: 100%██ 112/112 [00:00<?, ?B/s]
config.json: 100%██ 190/190 [00:00<00:00, 206kB/s]
Embedding model loaded
Dimension: 384
First eight values: [0.0008187148487195373, 0.005221615545451641, -0.038859017193
31741, -0.018897101283073425, 0.0531141459941864, 0.03502210974693298, -0.0529872
7750778198, 0.15177902579307556]
PS D:\data260-3209> █
```

Corpus

Code snippet:

```
5
6
7  def load_corpus_documents():
8      documents = []
9
10     for file_path in sorted(CORPUS_DIR.glob("*.json")):
11         data = json.loads(file_path.read_text(encoding="utf-8"))
12
13         for study in data.get("studies", []):
14             study_text = json.dumps(
15                 study,
16                 ensure_ascii=False,
17                 indent=2,
18             )
19
20             documents.append(
21                 Document(
22                     text=study_text,
23                     metadata={"source": f"corpus/{file_path.name}"},
24                 )
25             )
26
27     return documents
```

```
clinicaltrials_breast_cancer.json 1692826
clinicaltrials_covid19.json       4415388
clinicaltrials_diabetes.json      1856056
```

```
PS D:\data260-3209> (Get-ChildItem corpus | Measure-Object -Property Length -Sum).Sum
7964270
```

```
PS D:\data260-3209> Get-FileHash corpus\*.json -Algorithm SHA256
```

Algorithm	Hash
-----------	------

Three local ClinicalTrials.gov snapshots were loaded. The corpus contains 300 study records and is 7964270 bytes, which is greater than the required 200 KB. Each study keeps its source filename in metadata.

Source:

The clinical-trial records were downloaded from the public ClinicalTrials.gov Data API.

```

PS D:\data260-3209> Invoke-WebRequest -Uri "https://clinicaltrials.gov/api/v2/studies?query.cond=diabetes&pageSize=100&format=json" -OutFile "corpus\clinicaltrials_diabetes.json"
>>
>> Invoke-WebRequest -Uri "https://clinicaltrials.gov/api/v2/studies?query.cond=breast%20cancer&format=json" -OutFile "corpus\clinicaltrials_breast_cancer.json"
>>
>> Invoke-WebRequest -Uri "https://clinicaltrials.gov/api/v2/studies?query.cond=covid-19&format=json" -OutFile "corpus\clinicaltrials_covid19.json"
PS D:\data260-3209> Get-ChildItem corpus | Select-Object Name,Length

Name                                     Length
----                                     -
clinicaltrials_breast_cancer.json      1692826
clinicaltrials_covid19.json            4415388
clinicaltrials_diabetes.json           1856056

PS D:\data260-3209> (Get-ChildItem corpus | Measure-Object -Property Length -Sum).Sum
7964270

```

Questions

Five clinical-trial questions were written in questions.yaml before retrieval results were generated. Each question includes an expected answer and expected source file. The same five questions will be tested with all three chunking techniques.

```

auth.py X SOURCES.md X CORPUS_MANIFEST.json X ! questions.yaml X retrieve_hw03.py U X home.html X login.html
reports > hw03 > ! questions.yaml
1 - id: q1
2   question: "What was the overall study status of NCT00741390, the study evaluating blood volume and perceived pain in
3   expected_answer: "COMPLETED"
4   expected_source: "corpus/clinicaltrials_diabetes.json"
5
6 - id: q2
7   question: "What was the overall study status of NCT04144023, the vaccine study before surgery for HER2-expressing c
8   expected_answer: "RECRUITING"
9   expected_source: "corpus/clinicaltrials_breast_cancer.json"
10
11 - id: q3
12   question: "What was the overall study status of NCT04375410, the study about intensified handwashing and hand disin
13   expected_answer: "COMPLETED"
14   expected_source: "corpus/clinicaltrials_covid19.json"
15
16 - id: q4
17   question: "Which drug is named in the title of NCT03704818, a study about effects on hypoglycemia?"
18   expected_answer: "Dapagliflozin"
19   expected_source: "corpus/clinicaltrials_diabetes.json"
20
21 - id: q5
22   question: "What type of treatment or procedure is studied in NCT04817813, which examines long-term outcomes after b
23   expected_answer: "Liver metastasis surgery"
24   expected_source: "corpus/clinicaltrials_breast_cancer.json"

```

Token-based chunking

Code snippet:

```
66
67
68  ✓ def build_token_nodes(documents):
69  ✓     splitter = TokenTextSplitter(
70      |         chunk_size=256,
71      |         chunk_overlap=40,
72      |     )
73      |     return splitter.get_nodes_from_documents(documents)
74
75
```

Token-based chunking divides the documents into fixed-size chunks of 256 tokens. An overlap of 40 tokens preserves context between neighboring chunks.

Semantic chunking

```
✓ def build_semantic_nodes(documents, embed_model):
✓     splitter = SemanticSplitterNodeParser(
    |         buffer_size=1,
    |         breakpoint_percentile_threshold=95,
    |         embed_model=embed_model,
    |     )
    |
    |     return splitter.get_nodes_from_documents(documents)
```

Semantic chunking uses embeddings to identify meaningful topic boundaries. A buffer size of 1 and a breakpoint percentile of 95 were used.

Sentenced Chunking


```

6  def build_in_memory_index(nodes, embed_model):
7      vector_store = SimpleVectorStore()
8      storage_context = StorageContext.from_defaults(
9          vector_store=vector_store
10     )
11
12     return VectorStoreIndex(
13         nodes=nodes,
14         storage_context=storage_context,
15         embed_model=embed_model,
16         show_progress=False,
17     )

```

After chunking, we built a searchable vector index for each of the three techniques. The index stores the embedding of every chunk so that we can later search it with a question and find the most similar chunks.

```

5
6
7  def cosine_similarity(vector_a, vector_b):
8      vector_a = np.asarray(vector_a, dtype=float)
9      vector_b = np.asarray(vector_b, dtype=float)
10
11     denominator = np.linalg.norm(vector_a) * np.linalg.norm(vector_b)
12
13     if denominator == 0:
14         return 0.0
15
16     return float(np.dot(vector_a, vector_b) / denominator)
17
18

```

This function takes a question, converts it into an embedding, and searches the index for the top 3 most similar chunks. For each result it records the similarity score, the cosine similarity, the chunk length, a text preview, and the retrieval time in milliseconds.

```

=== Token | q1 ===
Query: What was the overall study status of NCT00741390?
Query embedding dimension: 384
First eight query values: [-0.0312348622828722, 0.03041848912835121, -0.026638036593794823, 0.0783847644925
1175, -0.04302878677845001, -0.001879774034023285, -0.09899290651082993, 0.12798288464546204]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 95.20 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.4359638881807629 | 0.319042 | 913 | provides the quantitative measurement of where the patient is at
any point in time during their course of treatment and how they compare with other patients. 6.
2 | 0.42549320660145357 | 0.256901 | 797 | virus at Screening. 21. The participant has donated or received
any blood products within 12 weeks prior to Screening. 22. The participant received medication p
3 | 0.4244457665940457 | 0.241834 | 559 | * Definitively treated carcinoma-in-situ of the cervix (Stage 0 c
ervical cancer) 17. Treatment or prophylaxis of COVID-19 with a monoclonal antibody product or

=== Semantic | q1 ===
Query: What was the overall study status of NCT00741390?
Query embedding dimension: 384
First eight query values: [-0.0312348622828722, 0.03041848912835121, -0.026638036593794823, 0.0783847644925
1175, -0.04302878677845001, -0.001879774034023285, -0.09899290651082993, 0.12798288464546204]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 62.01 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.5188553533588387 | 0.452245 | 32 | The study will comprise of: 1.
2 | 0.42903456473547447 | 0.225834 | 175 | Participant who are participating in other clinical trials or wh
o have participated in other clinical trials within 3 months. 4. Cases of severe shock and respi
3 | 0.42232241434809226 | 0.336942 | 68 | * Plan to remain living in the area for the duration of the study
.

=== Sentence-window | q1 ===
Query: What was the overall study status of NCT00741390?
Query embedding dimension: 384
First eight query values: [-0.0312348622828722, 0.03041848912835121, -0.026638036593794823, 0.0783847644925
1175, -0.04302878677845001, -0.001879774034023285, -0.09899290651082993, 0.12798288464546204]

```

```

1175, -0.04302878677845001, -0.001879774034023285, -0.09899290651082993, 0.12798288464546204]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 309.85 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.5414649978090086 | 0.426321 | 61 | Test type and result must be documented in the study record.
2 | 0.5374898056392216 | 0.290464 | 91 | Tests to be evaluated are to be determined, but this will depend o
n availability and cost.
3 | 0.5272142485939854 | 0.294982 | 162 | Be able to understand and agrees to fully comply with defined and
described study procedures and be available for all study visits for the entire study duration
PS D:\data260-3209>
PS D:\data260-3209> py -3 12_reports\bw03\scripts\retrieve_bw03.py

```

In the q1 results, the three retrieved chunks had moderate-to-high similarity scores, but none of them contained the correct study status for NCT00741390. This shows an important limitation of embedding-based retrieval: the similarity score measures how related the meaning is, not whether the chunk actually contains the correct factual answer. A chunk can look similar to the question and still be the wrong answer.

All 15 Runs Summary:

Each question was executed with all three techniques. The retrieval function recorded the query vector information, document-vector shapes, similarity scores, chunk lengths, previews, and retrieval latency. The complete machine-readable outputs for all 15 runs are stored in reports/hw03/raw/.

```
=== Token | q1 ===
Query: What was the overall study status of NCT00741390, the study evaluating blood volume and perceived pain during fingerstick blood glucose monitoring?
Query embedding dimension: 384
First eight query values: [-0.07211301475763321, -0.024431223049759865, -0.06613186746835709, 0.05610841140151024, -0.05628875643014908, -0.05587322264909744, 0.004312762524932623, 0.15470018982887268]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 85.79 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7038398327314511 | 0.729609 | 1179 | Study ID: NCT00741390 Overall status: COMPLETED Evaluation of Blood Volume and Perceived Pain During Fingerstick Monitoring of Blood Glucose When Using Differ
2 | 0.6322120490035178 | 0.605481 | 990 | evaluate in Visit 2 (pain during lancing). The lancet/lancing device combinations assigned to each arm are shown in the Assigned Interventions Table below. Con
3 | 0.6173197573512237 | 0.580401 | 1064 | 1hour 1H30, 2H, 2H30 and 3H post meal. Blood glucose will be measured with three different techniques: venous blood glucose measured with a COBAS 8000 automatic
Saved: D:\data260-3209\reports\hw03\raw\token_q1.json

=== Token | q2 ===
Query: What was the overall study status of NCT04144023, the vaccine study before surgery for HER2-expressing ductal carcinoma in situ?
Query embedding dimension: 384
First eight query values: [0.0032286858186125755, -0.0101593732838623, -0.017701147124171257, -0.0026005965191870928, -0.04276636987924576, -0.034145087003707886, -0.06255311518907547, 0.14052920043468475]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 89.25 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6571503708262523 | 0.631476 | 927 | macrophage-colony-stimulating factor (GM-CSF) admixed with multi-epitope HER2 peptide vaccine H2NVAC intradermally on day 1 of each cycle. Treatment repeats eve
2 | 0.6494491876986211 | 0.699712 | 990 | Study ID: NCT04144023 Overall status: RECRUITING A Phase IB Trial of Neoadjuvant Multi-Epitope HER2 Peptide Vaccine in Patients With HER2-Expressing DCIS Thi
3 | 0.6242815650295264 | 0.553997 | 1067 | receptor (AdHER)/neu dendritic cell vaccine, is custom-made using an individual's own immune cells. These cells will be collected and used to produce the vaccine
Saved: D:\data260-3209\reports\hw03\raw\token_q2.json

=== Token | q3 ===
Query: What was the overall study status of NCT04375410, the study about intensified handwashing and hand disinfection among children during the COVID-19 pandemic?
Query embedding dimension: 384
First eight query values: [-0.05497710779309273, 0.028032172471284866, 0.014938456006348133, 0.02508869208395481, 0.03670467808842659, -0.014153738506138325, -0.006509262602776289, 0.11421902477741241]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 91.99 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.726752373053149 | 0.699245 | 1200 | Study ID: NCT04375410 Overall status: COMPLETED Re-opening of the Country and Dermal Consequences of Mandatory Intensified Handwash and Hand Disinfection Among
2 | 0.6470781853920774 | 0.606614 | 809 | The study group wished to investigate whether this has had implications on symptoms of dry, red, itchy or sore hand
```

3	0.5601038670554558	0.519246	2414	Study ID: NCT05319496	Overall status: COMPLETED	Intermittently Scanned CGM Versus Usual Care With Diabetes Educa
---	--------------------	----------	------	-----------------------	---------------------------	--

```

=== Semantic | q2 ===
Query: What was the overall study status of NCT04144023, the vaccine study before surgery for HER2-expressing ductal carcinoma in situ?
Query embedding dimension: 384
First eight query values: [0.0032286858186125755, -0.01015937328338623, -0.017701147124171257, -0.0026005965191870928, -0.04276636987924576, -0.034145087003707886, -0.06255311518907547, 0.14052920043468475]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 74.91 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6494491876986211 | 0.693383 | 6046 | Study ID: NCT04144023 Overall status: RECRUITING A Phase IB Trial of Neoadjuvant Multi-Epitope HER2 Peptide Vaccine in Patients With HER2-Expressing DCIS This
2 | 0.6317916065166809 | 0.474835 | 163 | * Diagnosed with or treated for cancer within the previous 2 years, other than breast cancer or non-melanoma carcinoma of the skin. * Pregnant and nursing female
3 | 0.6083434145011762 | 0.353387 | 99 | * Participation in another clinical study with an investigational product during the last 3 months.
Saved: D:\data260-3209\reports\hw03\raw\semantic_q2.json

=== Semantic | q3 ===
Query: What was the overall study status of NCT04375410, the study about intensified handwashing and hand disinfection among children during the COVID-19 pandemic?
Query embedding dimension: 384
First eight query values: [-0.05497710779309273, 0.028032172471284866, 0.014938456006348133, 0.02508869208395481, 0.03670467808842659, -0.014153738506138325, -0.006509262602776289, 0.11421902477741241]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 69.26 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.743610568000066 | 0.744016 | 714 | Study ID: NCT04375410 Overall status: COMPLETED Re-opening of the Country and Dermal Consequences of Mandatory Intensified Handwash and Hand Disinfection Among
2 | 0.6630460934595264 | 0.640649 | 1103 | During the COVID-19 pandemic Denmark closed down, including schools, kindergartens and daycare. After reopening of the country 5 weeks later obligatory frequent
3 | 0.5702501605138209 | 0.554952 | 658 | Study ID: NCT05234359 Overall status: COMPLETED Rapid Research in the CHILD Cohort to Inform Canada's Response to the COVID-19 Pandemic: Investigating the Pre
Saved: D:\data260-3209\reports\hw03\raw\semantic_q3.json

=== Semantic | q4 ===
Query: Which drug is named in the title of NCT03704818, a study about effects on hypoglycemia?
Query embedding dimension: 384
First eight query values: [6.961908366065472e-05, 0.03366794437170029, -0.07190270721912384, 0.06643209606409073, 0.0014114778023213148, -0.016972485929727554, 0.01590210385620594, 0.1514713615179062]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 74.41 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.5763795618907352 | 0.589858 | 787 | Study ID: NCT00488527 Overall status: COMPLETED Multicentric, Open Label Clinical Trial. Use of Optimal Method to Initiate and Maintain Lantus Therapy (Insulin)
2 | 0.5554968204500101 | 0.552186 | 3125 | Describe the adverse events Evaluate the safety of using the medication according to the incidence and relevance of the hypoglycemia events, (symptomatic, diurnal
3 | 0.5404205123813469 | 0.370403 | 415 | Sign an informed consent and assent. Exclusion Criteria: 1. The disease is progressing rapidly; 2. The patient is receiving therapy of other new drugs; 3. Ev

```

```

=== Semantic | q5 ===
Query: What type of treatment or procedure is studied in NCT04817813, which examines long-term outcomes after breast cancer liver metastasis?
Query embedding dimension: 384
First eight query values: [-0.01756357029080391, -0.01372707448899746, -0.019433744251728058, -0.04338538274168968, 0.00980775710195303, -0.03185320273041725, -0.10213389247655869, 0.0991542860865593]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 82.35 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7078329993715359 | 0.710502 | 3264 | Study ID: NCT04817813 Overall status: UNKNOWN LIBREAST STUDY: Long-term Outcomes After Breast Cancer Liver Metastasis Surgery: an European, Retrospective, Single-Arm Phase II Study * American Society of Anesthesiologists (ASA) score I-III
2 | 0.6695921763227933 | 0.632558 | 204 | * Scheduled surgery for breast cancer liver metastases between January 1st 2010 and December 31st 2015 * American Society of Anesthesiologists (ASA) score I-III
3 | 0.6440538672831712 | 0.562221 | 2895 | Each treatment cycle will last 3 weeks. Treating physician will select agent up to 41 patients in each cohort. Final primary endpoint analysis will use combined
Saved: D:\data260-3209\reports\hw03\raw\semantic_q5.json

=== Sentence-window | q1 ===
Query: What was the overall study status of NCT00741390, the study evaluating blood volume and perceived pain during fingerstick blood glucose monitoring?
Query embedding dimension: 384
First eight query values: [-0.07211301475763321, -0.024431223049759865, -0.06613186746035709, 0.05610841140151024, -0.05628875643014908, -0.05587322264909744, 0.004312762524932623, 0.15470018982887268]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 334.37 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7650619969347089 | 0.778408 | 358 | Study ID: NCT00741390 Overall status: COMPLETED Evaluation of Blood Volume and Perceived Pain During Fingerstick Monitoring of Blood Glucose When Using Different Techniques
2 | 0.6984533621837079 | 0.535264 | 259 | Blood glucose will be measured with three different techniques: venous blood glucose measured with a COBAS 8000 automatic analyzer, SMBG measured with a commercial
3 | 0.6478575408694449 | 0.431097 | 117 | A clinical examination will be performed, including height, weight, body mass index, blood pressure, and heart rate
Saved: D:\data260-3209\reports\hw03\raw\sentence_window_q1.json

=== Sentence-window | q2 ===
Query: What was the overall study status of NCT04144023, the vaccine study before surgery for HER2-expressing ductal carcinoma in situ?
Query embedding dimension: 384
First eight query values: [0.0032286858186125755, -0.01015937328338623, -0.017701147124171257, -0.0026005965191870928, -0.04276636987924576, -0.034145087003707886, -0.06255311518907547, 0.14052920043468475]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 355.07 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.739132925773091 | 0.789360 | 326 | Study ID: NCT04144023 Overall status: RECRUITING A Phase IB Trial of Neoadjuvant Multi-Epitope HER2 Peptide Vaccine in Patients With HER2-Expressing DCIS This
2 | 0.6996766847221723 | 0.631791 | 112 | Researchers are testing a therapeutic cancer vaccine designed to stimulate the immune system to recognize HER2.
3 | 0.6871305584828689 | 0.560202 | 154 | Patients treated at dose level 3 receive GM-CSF admixed with multi-epitope HER2 peptide vaccine H2NVAC intradermally on days 1, 4, 8, and 15 for 1 cycle.

```

```

=== Sentence-window | q3 ===
Query: What was the overall study status of NCT04375410, the study about intensified handwashing and hand disinfection among children during the COVID-19 pandemic?
Query embedding dimension: 384
First eight query values: [-0.05497710779309273, 0.028032172471284866, 0.014938456006348133, 0.02508869208395481, 0.03670467808842659, -0.014153738506138325, -0.006509262602776289, 0.11421902477741241]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 337.76 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.8571915265781284 | 0.848771 | 370 | Study ID: NCT04375410 Overall status: COMPLETED Re-opening of the Country and Dermal Consequences of Mandatory Intensified Handwash and Hand Disinfection Amon
2 | 0.7659219451385635 | 0.644167 | 94 | Also the study group wished to investigate if frequency or use of hand disinfection mattered.
3 | 0.6741543187212135 | 0.445515 | 198 | After reopening of the country 5 weeks later obligatory frequent handwash was a requirement from the government and routines has been implemented at all reopene
Saved: D:\data260-3209\reports\hw03\raw\sentence_window_q3.json

=== Sentence-window | q4 ===
Query: Which drug is named in the title of NCT03704818, a study about effects on hypoglycemia?
Query embedding dimension: 384
First eight query values: [6.961908366065472e-05, 0.03366794437170029, -0.07190270721912384, 0.06643209606409073, 0.0014114778023213148, -0.0169724859297275, 4, 0.01590210385620594, 0.1514713615179062]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 344.53 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7119353256340393 | 0.623080 | 185 | Describe the adverse events Evaluate the safety of using the medication according to the incidence and relevance of the hypoglycemia events, (symptomatic, diurn
2 | 0.6358510654057692 | 0.505253 | 59 | T1DM treatment with any other antihyperglycemic drug (e.g.
3 | 0.6332882281055691 | 0.504219 | 134 | In clinical trials, general safety was high, glycemic control increased and the incidence of hypoglycaemia was significantly reduced.
Saved: D:\data260-3209\reports\hw03\raw\sentence_window_q4.json

=== Sentence-window | q5 ===
Query: What type of treatment or procedure is studied in NCT04817813, which examines long-term outcomes after breast cancer liver metastasis?
Query embedding dimension: 384
First eight query values: [-0.01756357029080391, -0.01372707448899746, -0.019433744251728058, -0.04338538274168968, 0.00980775710195303, -0.0318532027304172, -0.10213389247655869, 0.0991542860865593]
Query vector shape: (384,)
Document vector shape: (3, 384)
Retrieval latency: 361.81 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7622454284750666 | 0.646497 | 224 | Surgery is becoming more practical and effective than conservative treatment in improving the outcomes of patients with breast cancer liver metastasis and liver
2 | 0.7525216180754207 | 0.670657 | 112 | The proper indication for surgical treatment of breast cancer liver metastasis is still a matter of discussion.
3 | 0.702087580460312 | 0.651907 | 162 | * Scheduled surgery for breast cancer liver metastases between January 1st 2010 and December 31st 2015 * American Society of Anesthesiologists (ASA) score I-III
Saved: D:\data260-3209\reports\hw03\raw\sentence_window_q5.json

```

In 315 Col 32 Spaces:4

Verify the count and valid JSON

```

15
PS D:\data260-3209> py -3.12 -c "import glob,json; files=glob.glob('reports/hw03/raw/*.json'); [json.load(open(f,encoding='utf-8')) for f in files]; print('Raw JSON files:', len(files)); print('All raw JSON files are valid!')"
Raw JSON files: 15
All raw JSON files are valid

```

Metrics comparison

Technique	Chunks	Avg chunk length	Top-1 cosine	Mean@k cosine	Recall@k	Mean retrieval latency (ms)
Token	1494	987.4	0.6939	0.6130	1.0000	103.30
Semantic	820	1690.5	0.6917	0.5808	1.0000	82.12
Sentence-window	7863	190.9	0.7421	0.6177	1.0000	346.71

Avg chunk length → averaged the per-question "Avg retrieved chunk length" values from your per-question table:

Token: $(1077.7 + 994.7 + 1064.3 + 605.0 + 1194.3) \div 5 = 987.4$

Semantic: $(1961.3 + 2102.7 + 825.0 + 1442.3 + 2121.0) \div 5 = 1690.5$

Sentence-window: $(244.7 + 197.3 + 220.7 + 126.0 + 166.0) \div 5 = 190.9$

```

2
3 ## Full-corpus chunk statistics
4
5 | Technique | Chunks | Average chunk length (characters) | Minimum length | Maximum length |
6 |---|---|---|---|---|
7 | Token | 1494 | 975.78 | 147 | 1427 |
8 | Semantic | 820 | 1515.05 | 3 | 12408 |
9 | Sentence-window | 7863 | 158.00 | 2 | 5351 |
0
1 ## Per-question retrieval results
2
3 | Question | Technique | Top-1 cosine | Mean@k cosine | Recall@k | Latency (ms) |
4 |---|---|---|---|---|
5 | q1 | Semantic | 0.7207 | 0.6108 | True | 109.70 |
6 | q2 | Semantic | 0.6934 | 0.5072 | True | 74.91 |
7 | q3 | Semantic | 0.7440 | 0.6465 | True | 69.26 |
8 | q4 | Semantic | 0.5899 | 0.5041 | True | 74.41 |
9 | q5 | Semantic | 0.7105 | 0.6351 | True | 82.35 |
0 | q1 | Sentence-window | 0.7784 | 0.5816 | True | 334.37 |
1 | q2 | Sentence-window | 0.7894 | 0.6605 | True | 355.07 |
2 | q3 | Sentence-window | 0.8488 | 0.6462 | True | 337.76 |
3 | q4 | Sentence-window | 0.6231 | 0.5442 | True | 344.53 |
4 | q5 | Sentence-window | 0.6707 | 0.6564 | True | 361.81 |
5 | q1 | Token | 0.7296 | 0.6385 | True | 85.79 |
6 | q2 | Token | 0.6997 | 0.6284 | True | 89.25 |
7 | q3 | Token | 0.6992 | 0.6108 | True | 91.99 |
8 | q4 | Token | 0.6215 | 0.5141 | True | 86.72 |
9 | q5 | Token | 0.7195 | 0.6733 | True | 162.74 |
0
1 ## Technique summary
2
3 | Technique | Chunks | Average chunk length | Mean top-1 cosine | Mean@k cosine | Recall@k | Mean latency (ms) |
4 |---|---|---|---|---|---|
5 | Token | 1494 | 975.78 | 0.6939 | 0.6130 | 1.0000 | 103.30 |
6 | Semantic | 820 | 1515.05 | 0.6917 | 0.5808 | 1.0000 | 82.12 |
7 | Sentence-window | 7863 | 158.00 | 0.7421 | 0.6177 | 1.0000 | 346.71 |
8
9 ## High-score incorrect retrieval analysis
0
1 For q4, the expected answer was Dapagliflozin from study NCT03704818.
2 The Token method ranked a different diabetes study, NCT00488527, first.
3 The incorrect result had a store score of 0.6117 and cosine similarity of 0.6215.
4 Its preview referred to Lantus therapy and did not contain Dapagliflozin.
5 The embedding likely considered it similar because both studies contain related
6 diabetes, medication, and hypoglycemia language. This demonstrates that
7 embedding similarity measures semantic relatedness, not exact answer correctness.
8

```

Q. Find at least one confidently scored retrieval that does not contain the answer. If your five questions don't produce one, add questions until you find one. Explain why the embedding likely considered it similar.

For q4, the expected answer was Dapagliflozin from study NCT03704818.

Question: Which drug is named in the title of NCT03704818, a study about effects on hypoglycemia?

- Expected answer: Dapagliflozin
- Token method's top result: Study ID: NCT00488527 Use of Optimal Method to Initiate and Maintain Lantus Therapy
- Scores: Store score: 0.6117, Cosine similarity: 0.6215

This is a confidently scored but incorrect retrieval because it does not contain NCT03704818, Dapagliflozin

Observation:

Sentence-window performed best overall for this corpus. It achieved the highest mean top-1 cosine similarity of 0.7421 and the highest mean@k cosine similarity of 0.6177. Its sentence-level chunks preserved nearby context, helping the retrieved text remain closely related to the question. However, this method created the most chunks, 7,863, and had the highest average retrieval latency of 346.71 ms.

Performance varied by question. Sentence-window produced the strongest top-1 similarity for q1 through q4, while Token performed better for q5 and had the strongest mean@k result for q1 and q5. Token also provided a better speed-quality balance with an average latency of 103.30 ms. Semantic chunking created the fewest chunks and had the lowest average latency, but its average similarity scores were lower than the other two techniques. No optional extra questions were used.

Conclusion:

For this clinical-trial corpus, Sentence-window chunking was the best overall technique because it achieved the highest mean top-1 cosine similarity of 0.7421 and the highest mean@k cosine similarity of 0.6177. Its sentence-level chunks preserved nearby context, which helped the retrieved text remain relevant to the questions, and it achieved perfect Recall@k of 1.0, like the other two techniques. However, Sentence-window created 7863 chunks and had the highest average retrieval latency at 346.71 ms, making it more computationally expensive. Token chunking was a practical alternative because it achieved strong retrieval quality with only 1494 chunks and a much lower average latency of 103.30 ms, while Semantic chunking was the fastest but produced lower average similarity scores.

Verification.JSON

The verification script passed all checks. It confirmed that the application routes work, the dashboard is protected, all 15 retrieval files are present, and the required metadata files exist.

```

34     "check": "All raw files contain required retrieval fields",
35     "pass": true,
36     "details": "15 of 15 files passed"
37   },
38   {
39     "check": "Required HW3 metadata files exist",
40     "pass": true,
41     "details": "All required files exist"
42   }
43 ],
44 "overall_pass": true
45 }

```

Problems Output Debug Console **Terminal** Ports

```

    "details": {
      "status": 303,
      "location": "/login"
    }
  },
  {
    "check": "Exactly 15 raw retrieval JSON files exist",
    "pass": true,
    "details": "Found 15 raw JSON files"
  },
  {
    "check": "All raw files contain required retrieval fields",
    "pass": true,
    "details": "15 of 15 files passed"
  },
  {
    "check": "Required HW3 metadata files exist",
    "pass": true,
    "details": "All required files exist"
  }
],
"overall_pass": true
}

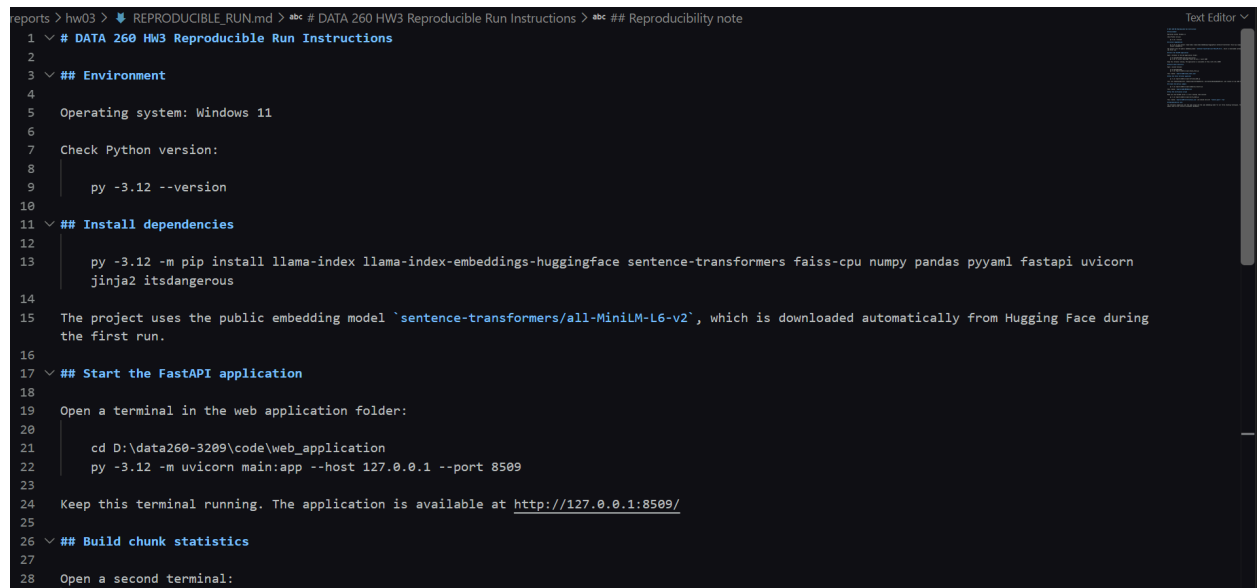
```

Verification written to: D:\data260-3209\reports\hw03\verification.json

PS D:\data260-3209>

Report explanation

Reproducible run instructions were included with the project. They document how to start the FastAPI application, generate chunk statistics, run all three retrieval pipelines, create the metrics summary, and run the verification script.



```
reports > hw03 > REPRODUCIBLE_RUN.md > # DATA 260 HW3 Reproducible Run Instructions > ## Reproducibility note
1  # DATA 260 HW3 Reproducible Run Instructions
2
3  ## Environment
4
5  Operating system: Windows 11
6
7  Check Python version:
8
9  py -3.12 --version
10
11 ## Install dependencies
12
13 py -3.12 -m pip install llama-index llama-index-embeddings-huggingface sentence-transformers faiss-cpu numpy pandas pyyaml fastapi uvicorn
14   jinja2 itsdangerous
15
16 The project uses the public embedding model 'sentence-transformers/all-MiniLM-L6-v2', which is downloaded automatically from Hugging Face during
17 the first run.
18
19 ## Start the FastAPI application
20
21 Open a terminal in the web application folder:
22
23 cd D:\data260-3209\code\web_application
24 py -3.12 -m uvicorn main:app --host 127.0.0.1 --port 8509
25
26 Keep this terminal running. The application is available at http://127.0.0.1:8509/
27
28 ## Build chunk statistics
29
30 Open a second terminal:
```

AI Use and Verification

1. How AI assistance was used

I used an AI assistant as a programming guide while completing HW3. It helped me understand the assignment requirements and provided code suggestions for creating `auth.py`, updating `main.py`, adding login and logout routes, managing sessions, protecting the dashboard, and applying Bootstrap styling to the HTML pages. For Part 2, it also suggested code for loading the clinical-trial documents, creating the three LlamaIndex chunking pipelines, generating embeddings, retrieving results, calculating cosine similarity, creating metrics, and running verification checks.

I did not use the assistant to submit pre-generated results. I entered and modified the suggested code in VS Code, installed the required packages, ran the programs locally, checked errors, tested the application, captured screenshots, reviewed the retrieval outputs, and created the final report. I also made the final decisions about which results and evidence to include.

2. Incorrect or unsuitable AI-produced output

The first retrieval implementation chunked the complete JSON records directly. This caused chunks to contain JSON syntax such as braces, field names and quoted values instead of clear clinical-trial text. The first retrieval results were also not useful for the question about the study status of NCT00741390.

3. How the problem was detected

I inspected the printed chunk previews and retrieval results. The previews contained fragmented JSON content and the top results for the status question did not clearly contain the expected study status. I compared the results with the expected answers in questions.yaml

4. What was changed and why

I changed the corpus loader to extract readable fields from each study, including the study title, summary, detailed description, conditions, eligibility criteria, study ID, and overall status. This created more meaningful chunks for embedding and retrieval. I then reran the three chunking techniques and regenerated the raw results and metrics. The final pipeline produced meaningful text previews and complete comparison results for all 15 question-technique combinations.